

Linear Algebra II (2026 Fall)

Studying Contents :

Chapter 6. Inner Product Spaces

- Section 6.1 : Inner Products and Norms

Chapter 6. Inner Product Spaces

Remark 1. : Let $V = \mathbb{R}^3$.

(1) For a vector $u = (a,b,c)$, we can define the length of u , denoted by $\|u\|$, by

$$\|u\| = \sqrt{a^2 + b^2 + c^2}.$$

(2) For two members $u = (a,b,c)$ and $v = (d,e,f)$ in $V = \mathbb{R}^3$,

we define the distance between u and v , denoted by $d(u,v)$, by

$$d(u,v) = \sqrt{(a-d)^2 + (b-e)^2 + (c-f)^2}.$$

Let θ be the angle between u and v . Then

$$d(u,v)^2 = \|u\|^2 + \|v\|^2 - 2\|u\| \|v\| \cos \theta.$$

This gives us $\cos \theta = \frac{ad + be + cf}{\|u\| \times \|v\|}$.

For $\langle u,v \rangle := ad + be + cf$, it holds that

$$\cos \theta = \frac{\langle u,v \rangle}{\|u\| \times \|v\|} \text{ and hence } \langle u,v \rangle = \|u\| \|v\| \cos \theta.$$

Remark 2. : Let $V = \mathbb{R}^n$.

(1) For two vectors $u = (a_1, \dots, a_n)$ and $v = (b_1, \dots, b_n)$, let θ be the angle between u and v .

$$\|u\| = \text{the length of } u = \sqrt{a_1^2 + \dots + a_n^2}$$

$$\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\| \|v\| \cos \theta$$

$$\cos \theta = \frac{a_1 b_1 + \dots + a_n b_n}{\|u\| \times \|v\|}$$

(2) Define $\langle u, v \rangle := a_1 b_1 + \dots + a_n b_n$. Thus $\langle u, u \rangle = \|u\|^2$ and $\langle v, v \rangle = \|v\|^2$. Also,

$$\cos \theta = \frac{\langle u, v \rangle}{\|u\| \times \|v\|}.$$

The main purpose of this chapter is

to introduce the idea of distance or length and angle into vector spaces via a much richer structure, the so-called inner product space structure.

Note : Throughout this chapter, we assume that a field F is either $\mathbb{R}$ or $\mathbb{C}$.

Also, for any $z \in F$, we denote by $\bar{z}$ the complex conjugation of z .

That is, if $z = x + iy$ ($x, y \in \mathbb{R}$) then $\bar{z} = x - iy$.

Section 6.1 Inner Products and Norms

In this section, we will define an inner product on a vector space over $\mathbb{R}$ or $\mathbb{C}$.

This will enable us to extend

many geometric notions such as angle, length, and perpendicularity in $\mathbb{R}^2$ and $\mathbb{R}^3$.

Definition : Let V be a vector space over a field F . An inner product on V is a function

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow F$$

which satisfies the following conditions:

- (a) $\langle x + z, y \rangle = \langle x, y \rangle + \langle z, y \rangle$.
- (b) $\langle cx, y \rangle = c \langle x, y \rangle$.
- (c) $\overline{\langle x, y \rangle} = \langle y, x \rangle$, where the bar denotes complex conjugation.
- (d) $\langle x, x \rangle > 0$ if $x \neq 0$.

Remark : (1) Conditions (a) and (b) says that the inner product is linear in the first

component. More generally, it holds that $\left\langle \sum_{i=1}^n a_i v_i, y \right\rangle = \sum_{i=1}^n a_i \langle v_i, y \rangle$.

(2) If $F = \mathbb{R}$, then condition (c) reduces to $\langle x, y \rangle = \langle y, x \rangle$.

Example 1. : Let $V = F^n$ (where F is either $\mathbb{R}$ or $\mathbb{C}$).

For $x = (a_1, a_2, \dots, a_n)$ and $y = (b_1, b_2, \dots, b_n)$ in V , define

$$\langle x, y \rangle = \sum_{i=1}^n a_i \overline{b_i}.$$

Then $\langle \cdot, \cdot \rangle$ on V is an inner product.

For example, condition (a) can be shown as follows, where $z = (c_1, c_2, \dots, c_n)$:

$$\begin{aligned} \langle x + z, y \rangle &= \sum_{i=1}^n (a_i + c_i) \overline{b_i} = \sum_{i=1}^n a_i \overline{b_i} + \sum_{i=1}^n c_i \overline{b_i} \\ &= \langle x, y \rangle + \langle z, y \rangle. \end{aligned}$$

Terminologies :

- (1) The inner product in [Example 1](#) is called **the standard inner product** on $V = F^n$.
- (2) Suppose that $F = \mathbb{R}$. Then the conjugations are not needed.

In this case, this standard inner product is usually called **the dot product**.

Also, the inner product of x and y is denoted by $x \cdot y$ instead of $\langle x, y \rangle$.

Remark : Let $\langle \cdot, \cdot \rangle : V \times V \rightarrow F$ be an inner product on an n -dimensional vector space V .

(1) Let $\beta = \{v_1, v_2, \dots, v_n\}$ be a basis for V .

Then for $v = a_1v_1 + \dots + a_nv_n$ and $w = b_1v_1 + \dots + b_nv_n$, we have

$$\begin{aligned} \langle v, w \rangle &= \sum_{1 \leq i, j \leq n} a_i \times \langle v_i, v_j \rangle \times \overline{b_j} = \sum_{i=1}^n a_i \times \left(\sum_{j=1}^n \langle v_i, v_j \rangle \times \overline{b_j} \right) \\ &= (a_1, a_2, \dots, a_n) \begin{pmatrix} \langle v_1, v_1 \rangle & \langle v_1, v_2 \rangle & \dots & \langle v_1, v_n \rangle \\ \langle v_2, v_1 \rangle & \langle v_2, v_2 \rangle & \dots & \langle v_2, v_n \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle v_n, v_1 \rangle & \langle v_n, v_2 \rangle & \dots & \langle v_n, v_n \rangle \end{pmatrix} \begin{pmatrix} \overline{b_1} \\ \overline{b_2} \\ \vdots \\ \overline{b_n} \end{pmatrix}. \end{aligned}$$

Thus for $A \in M_{n \times n}(F)$ such that $A_{ij} = \langle v_i, v_j \rangle$, then we have $\langle v, w \rangle = [v]_{\beta}^t A [w]_{\beta}$.

(2) $A_{ij} = \langle v_i, v_j \rangle = \overline{\langle v_j, v_i \rangle} = \overline{A_{ji}}$ for all $1 \leq i, j \leq n$.

In particular, if $F = \mathbb{R}$ then A is symmetric since $A_{ij} = A_{ji}$ for all $1 \leq i, j \leq n$.

(3) A must be an invertible matrix.

Indeed, if $A[v]_{\beta} = 0$ then we have $\langle v, v \rangle = [v]_{\beta}^t A [v]_{\beta} = 0$ and hence $v = 0_V$.

Example 1. (Revisit) : Let $V = F^n$ (where F is either $\mathbb{R}$ or $\mathbb{C}$).

For $x = (a_1, a_2, \dots, a_n)$ and $y = (b_1, b_2, \dots, b_n)$ in V , define

$$\langle x, y \rangle = \sum_{i=1}^n a_i \overline{b_i}.$$

For the standard basis $\beta = \{e_1, e_2, \dots, e_n\}$ of V , we have

$$\langle e_i, e_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \text{ and} \\ 0 & \text{if } i \neq j. \end{cases}$$

Therefore the matrix A in the previous remark is the identity matrix I_n .

That is, if $x = (a_1, a_2, \dots, a_n)$ and $y = (b_1, b_2, \dots, b_n)$, then

$$\langle x, y \rangle = (a_1, a_2, \dots, a_n) I_n \begin{pmatrix} \overline{b_1} \\ \overline{b_2} \\ \vdots \\ \overline{b_n} \end{pmatrix}.$$

Example 2. : Let $\langle \cdot, \cdot \rangle : V \times V \rightarrow F$ be an inner product on a vector space V over F .

(1) For any $r > 0$, the function

$$\langle \cdot, \cdot \rangle' : V \times V \rightarrow F \text{ defined by } \langle x, y \rangle' = r \langle x, y \rangle$$

is another inner product on V .

Suppose that V is of dimension n , and let $\beta = \{v_1, v_2, \dots, v_n\}$ be a basis for V .

Also, let $A \in M_{n \times n}(F)$ such that $A_{ij} = \langle v_i, v_j \rangle$.

Then

$$\langle x, y \rangle' = r \langle x, y \rangle = r \times [x]_{\beta}^t A \overline{[y]_{\beta}} = [x]_{\beta}^t (rA) \overline{[y]_{\beta}}.$$

Thus the matrix corresponding to $\langle \cdot, \cdot \rangle'$ and β is rA .

(2) If $r \leq 0$, then the function

$$\langle \cdot, \cdot \rangle' : V \times V \rightarrow F \text{ defined by } \langle x, y \rangle' = r \langle x, y \rangle$$

fails to be an inner product on V , since condition (d) would not hold.

Example 3. : Let

$$V = C([0,1]) := \{ f : [0,1] \rightarrow \mathbb{R} \mid f \text{ is continuous} \}$$

be the vector space of real-valued continuous functions on $[0,1]$.

For $f, g \in V$, define $\langle f, g \rangle := \int_0^1 f(t)g(t)dt$.

Then $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is an inner product on V .

$$\begin{aligned} (\because) \text{ (a) } \langle f_1 + f_2, g \rangle &= \int_0^1 (f_1(t) + f_2(t))g(t)dt \\ &= \int_0^1 f_1(t)g(t)dt + \int_0^1 f_2(t)g(t)dt = \langle f_1, g \rangle + \langle f_2, g \rangle. \end{aligned}$$

$$\text{(b) } \langle cf, g \rangle = \int_0^1 (cf(t))g(t)dt = c \int_0^1 f(t)g(t)dt = c\langle f, g \rangle$$

$$\text{(c) } \langle g, f \rangle = \int_0^1 g(t)f(t)dt = \int_0^1 f(t)g(t)dt = \langle f, g \rangle = \overline{\langle f, g \rangle}$$

$$\text{(d) } \langle f, f \rangle = \int_0^1 f(t)^2 dt \geq 0, \text{ and } \langle f, f \rangle = 0 \text{ if and only if } f = 0.$$

Example : Let $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$ be an inner product on a vector space V over $\mathbb{C}$.

Then determine whether the following map defines an inner product on V :

$$\langle \cdot, \cdot \rangle' : V \times V \rightarrow \mathbb{C} \text{ defined by } \langle x, y \rangle' = \overline{\langle x, y \rangle}$$

$$\begin{aligned} \text{(a)} \quad \langle v_1 + v_2, w \rangle' &= \overline{\langle v_1 + v_2, w \rangle} = \overline{\langle v_1, w \rangle + \langle v_2, w \rangle} \\ &= \overline{\langle v_1, w \rangle} + \overline{\langle v_2, w \rangle} = \langle v_1, w \rangle' + \langle v_2, w \rangle' \end{aligned}$$

$$\text{(b)} \quad \langle cv, w \rangle' = \overline{\langle cv, w \rangle} = \overline{c \langle v, w \rangle} = \bar{c} \overline{\langle v, w \rangle} = \bar{c} \langle v, w \rangle'$$

$$\text{(c)} \quad \langle w, v \rangle' = \overline{\langle w, v \rangle} = \langle v, w \rangle = \overline{\langle v, w \rangle}'$$

$$\text{(d)} \quad \langle v, v \rangle' = \overline{\langle v, v \rangle} = \langle v, v \rangle > 0 \text{ whenever } v \neq 0_V.$$

In consequence, $\langle \cdot, \cdot \rangle'$ is not an inner product on V

since it satisfies (a), (c) and (d) but fails to satisfy (b).

Definition : For $A \in M_{m \times n}(F)$, we define the **conjugate transpose** (or **adjoint**) of A to be the $n \times m$ matrix A^* such that $(A^*)_{ij} = \overline{A_{ji}}$ for all i, j .

Example 4. :

$$\text{For } A = \begin{pmatrix} i & 1+2i \\ 2 & 3+4i \\ 3i & 2 \end{pmatrix}, \text{ we have } A^* = \begin{pmatrix} -i & 2 & -3i \\ 1-2i & 3-4i & 2 \end{pmatrix}.$$

Remark : Let $V = F^n$ (where F is either $\mathbb{R}$ or $\mathbb{C}$).

$$\text{For } x = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \text{ and } y = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \text{ in } V,$$

$$\langle x, y \rangle = \sum_{i=1}^n a_i \overline{b_i} \text{ is equal to } y^* x$$

Example 5. : Let $V = M_{n \times n}(F)$, and define

$$\langle A, B \rangle = \text{tr}(B^* A) \text{ for } A, B \in V.$$

Thus, if $A = [u_1 \ u_2 \ \cdots \ u_n]$ and $B = [v_1 \ v_2 \ \cdots \ v_n]$, then

$$B^* A = \begin{bmatrix} v_1^* \\ \vdots \\ v_n^* \end{bmatrix} [u_1 \ \cdots \ u_n] \text{ and hence } \text{tr}(B^* A) = \sum_{i=1}^n (B^* A)_{ii} = \sum_{i=1}^n v_i^* u_i.$$

Then $\langle \cdot, \cdot \rangle : V \times V \rightarrow F$ is an inner product on V .

For example, let $A, B, C \in V$. Then

$$\begin{aligned} \langle A + B, C \rangle &= \text{tr}(C^*(A + B)) = \text{tr}(C^* A + C^* B) \\ &= \text{tr}(C^* A) + \text{tr}(C^* B) = \langle A, C \rangle + \langle B, C \rangle. \end{aligned}$$

Also,

$$\begin{aligned} \langle A, A \rangle &= \text{tr}(A^* A) = \sum_{i=1}^n (A^* A)_{ii} = \sum_{i=1}^n \sum_{k=1}^n (A^*)_{ik} A_{ki} \\ &= \sum_{i=1}^n \sum_{k=1}^n \overline{A_{ki}} A_{ki} = \sum_{i=1}^n \sum_{k=1}^n |A_{ki}|^2. \end{aligned}$$

This is called the **Frobenius inner product**.

Example 5. (Continued) : Let $V = M_{n \times n}(F)$, and define

$$\langle A, B \rangle = \text{tr}(B^* A) \text{ for } A, B \in V.$$

Consider the standard basis $\beta = \{ E_{ij} \mid 1 \leq i, j \leq n \}$ for $V = M_{n \times n}(F)$.

Then it holds that

$$\langle E_{ij}, E_{kl} \rangle = E_{kl}^* E_{ij} = E_{\ell k} E_{ij} = \delta_{ik} \times \delta_{j\ell} = \begin{cases} 1 & \text{if } (i, j) = (k, \ell), \text{ and} \\ 0 & \text{if } (i, j) \neq (k, \ell). \end{cases}$$

Therefore, we have

$$\langle A, B \rangle = [A]_{\beta}^t I_{n^2} \overline{[B]_{\beta}}.$$

That is, if we regard $V = M_{n \times n}(F)$ as F^{n^2} in a standard way, then

the inner product $\langle A, B \rangle = \text{tr}(B^* A)$ on V

coincides with

the standard inner product on F^{n^2} .

Some Terminologies and Remark :

(1) A vector space V over F with a specific inner product $\langle \cdot, \cdot \rangle : V \times V \rightarrow F$ is called an **inner product space**.

(2) When $F = \mathbb{C}$, we call V a **complex inner product space**.

When $F = \mathbb{R}$, we call V a **real inner product space**.

(3) Let V be an inner product space by $\langle \cdot, \cdot \rangle : V \times V \rightarrow F$.

If W is a subspace of V , the same function $\langle x, y \rangle$ restricted to the vectors $x, y \in W$ makes W to be an inner product space.

Example : On $V = C([-1, 1])$, the two functions

$\langle f, g \rangle_1 := \int_0^1 f(t)g(t)dt$ and $\langle f, g \rangle_2 := \int_{-1}^1 f(t)g(t)dt$ define two distinct inner products.

For instance, if $f(x) = x$ and $g(x) = x^2$ then

$$\langle f, g \rangle_1 = \frac{1}{4} \text{ and } \langle f, g \rangle_2 = 0.$$

The above two inner products define two distinct inner products on V .

Theorem 6.1. : Let V be an inner product space.

Then for $x, y, z \in V$ and $c \in F$, the following statements are true.

$$(a) \quad \langle x, y+z \rangle = \langle x, y \rangle + \langle x, z \rangle.$$

$$(b) \quad \langle x, cy \rangle = \bar{c} \langle x, y \rangle.$$

$$(c) \quad \langle x, O_V \rangle = \langle O_V, x \rangle = 0.$$

$$(d) \quad \langle x, x \rangle = 0 \text{ if and only if } x = O_V.$$

$$(e) \quad \text{If } \langle x, y \rangle = \langle x, z \rangle \text{ for all } x \in V, \text{ then } y = z.$$

Proof : (a) We have

$$\begin{aligned} \langle x, y+z \rangle &= \overline{\langle y+z, x \rangle} = \overline{\langle y, x \rangle + \langle z, x \rangle} \\ &= \overline{\langle y, x \rangle} + \overline{\langle z, x \rangle} = \langle x, y \rangle + \langle x, z \rangle. \end{aligned}$$

(b), (c) : Obvious.

(d) $\langle x, x \rangle > 0$ by the definition of inner product space.

On the other hand, $\langle x, x \rangle = 0$ if $x = O_V$.

(e) If $\langle x, y \rangle = \langle x, z \rangle$ and so $\langle x, y-z \rangle = 0$ for all $x \in V$, then $\langle y-z, y-z \rangle = 0$.

Then $y-z = O_V$ and hence $y = z$. ■

Definition : Let V be an inner product space. For $x \in V$,

we define the norm of length of x by $\|x\| = \sqrt{\langle x, x \rangle}$.

Example 6. : Let $V = F^n$ be with the standard inner product.

If $x = (a_1, a_2, \dots, a_n)$, then $\|x\| = \|(a_1, a_2, \dots, a_n)\| = \sqrt{a_1^* a_1 + a_2^* a_2 + \dots + a_n^* a_n}$.

Thus

the norm about the standard inner product is the Euclidean definition of length.

Next, we will see that the well-known properties of Euclidean length in $\mathbb{R}^3$ hold in general.

Theorem 6.2. : Let V be an inner product space of F .

Then for all $x, y \in V$ and $c \in F$, the following statements are true.

(a) $\|cx\| = |c| \cdot \|x\|$.

(b) $\|x\| \geq 0$, and $\|x\| = 0$ if and only if $x = 0_V$.

(c) (Cauchy-Schwarz Inequality) $|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$.

(d) (Triangle Inequality) $\|x + y\| \leq \|x\| + \|y\|$.

Proof : (a), (b) Obvious.

(c) The result is obviously true if $y = O_V$.

Suppose that $y \neq O_V$ and hence $\langle y, y \rangle \neq 0$.

For any $c \in F$, we have

$$\begin{aligned} 0 \leq \|x - cy\|^2 &= \langle x - cy, x - cy \rangle = \langle x, x - cy \rangle - c \langle y, x - cy \rangle \\ &= \langle x, x \rangle - \bar{c} \langle x, y \rangle - c \langle y, x \rangle + c \bar{c} \langle y, y \rangle \\ &= \langle x, x \rangle - \bar{c} \langle x, y \rangle - c \overline{\langle x, y \rangle} + c \bar{c} \langle y, y \rangle \end{aligned}$$

For $c = \frac{\langle x, y \rangle}{\langle y, y \rangle}$, we get

$$\begin{aligned} &\langle x, x \rangle - \bar{c} \langle x, y \rangle - c \overline{\langle x, y \rangle} + c \bar{c} \langle y, y \rangle \\ &= \langle x, x \rangle - \frac{\overline{\langle x, y \rangle}}{\langle y, y \rangle} \langle x, y \rangle - \frac{\langle x, y \rangle}{\langle y, y \rangle} \overline{\langle x, y \rangle} + \frac{\langle x, y \rangle}{\langle y, y \rangle} \frac{\overline{\langle x, y \rangle}}{\langle y, y \rangle} \langle y, y \rangle \\ &= \langle x, x \rangle - \frac{|\langle x, y \rangle|^2}{\langle y, y \rangle} \geq 0. \end{aligned}$$

In consequence, it holds that

$$|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle \text{ and hence } |\langle x, y \rangle|^2 \leq \|x\|^2 \cdot \|y\|^2.$$

This completes the proof of the inequality $|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$.

(d) We have

$$\begin{aligned}
 \|x + y\|^2 &= \langle x + y, x + y \rangle = \langle x, x \rangle + \langle y, x \rangle + \langle x, y \rangle + \langle y, y \rangle \\
 &= \|x\|^2 + \|y\|^2 + (\langle x, y \rangle + \overline{\langle x, y \rangle}) \quad \text{Write } \langle x, y \rangle = \alpha + i\beta. \\
 &= \|x\|^2 + \|y\|^2 + 2\alpha \\
 &\leq \|x\|^2 + \|y\|^2 + 2\sqrt{\alpha^2 + \beta^2} \\
 &= \|x\|^2 + \|y\|^2 + 2|\langle x, y \rangle| \\
 &\leq \|x\|^2 + \|y\|^2 + 2\|x\| \cdot \|y\| = (\|x\| + \|y\|)^2.
 \end{aligned}$$

This proves the desired inequality $\|x + y\| \leq \|x\| + \|y\|$. ■

Example 7. : Let $V = \mathbb{R}^n$ be the standard inner product space.

Then $x = (a_1, a_2, \dots, a_n)$ and $y = (b_1, b_2, \dots, b_n)$, then

Cauchy-Schwarz Inequality and **Triangle Inequality** can be rewritten as follows:

$$\left| \sum_{i=1}^n a_i \bar{b}_i \right| \leq \left[\sum_{i=1}^n |a_i|^2 \right]^{1/2} \left[\sum_{i=1}^n |b_i|^2 \right]^{1/2} \quad \text{and} \quad \left[\sum_{i=1}^n |a_i + b_i|^2 \right]^{1/2} \leq \left[\sum_{i=1}^n |a_i|^2 \right]^{1/2} + \left[\sum_{i=1}^n |b_i|^2 \right]^{1/2}$$

Recall : For x and y in $\mathbb{R}^2$ or $\mathbb{R}^3$, we have

$$\langle x, y \rangle = \|x\| \cdot \|y\| \cos \theta$$

where θ ($0 \leq \theta \leq \pi$) denotes the angle between x and y .

In particular, if x and y are nonzero then

$$\langle x, y \rangle = 0 \text{ if and only if } \theta = \frac{\pi}{2}$$

if and only if x and y are perpendicular.

In the next definition,

we generalize the notion of perpendicularity to arbitrary inner product space.

Definition : Let V be an inner product space, and let x and y be members of V .

(1) We say that x and y are orthogonal (perpendicular) if $\langle x, y \rangle = 0$.

More generally,

a subset S of V is said to be orthogonal if any two distinct vectors in S are orthogonal.

(2) We say that x is a unit vector if $\|x\| = 1$.

A subset S of V is said to be orthonormal if

it is orthogonal and consists entirely of unit vectors.

Remark : Let $S = \{v_1, v_2, \dots\}$ be a subset of V .

(1) Let δ_{ij} denote the Kronecker delta. Then

S is orthonormal if and only if $\langle v_i, v_j \rangle = \delta_{ij}$ for all i, j .

(2) Let $c_1, c_2, \dots$ be nonzero scalars. Then

$S = \{v_1, v_2, \dots\}$ is orthogonal if and only if $S' = \{c_1 v_1, c_2 v_2, \dots\}$ is orthogonal.

(3) If $S = \{v_1, v_2, \dots\}$ is orthogonal, then the set $\bar{S} = \left\{ \frac{1}{\|v_1\|} v_1, \frac{1}{\|v_2\|} v_2, \dots \right\}$ is orthonormal.

This process of multiplying a nonzero vector by the reciprocal of its length

is called normalizing.

Example 8. : The subset $S = \{u = (1, 1, 0), v = (1, -1, 1), w = (-1, 1, 2)\}$ of F^3 is orthogonal.

Also $\|u\| = \sqrt{2}$, $\|v\| = \sqrt{3}$ and $\|w\| = \sqrt{6}$, and hence S is not orthonormal.

By normalizing process, we obtain the following orthonormal set:

$$\bar{S} = \left\{ \frac{1}{\sqrt{2}}(1, 1, 0), \frac{1}{\sqrt{3}}(1, -1, 1), \frac{1}{\sqrt{6}}(-1, 1, 2) \right\}$$

Let $H := \{ f : [0, 2\pi] \rightarrow \mathbb{C} \mid f \text{ is continuous} \}$ be

the set of all continuous complex-valued functions on the interval $[0, 2\pi]$.

This is a vector space over $\mathbb{C}$. Also every $f \in H$ can be written as

$f = f_1 + if_2$ where $f_1, f_2 : [0, 2\pi] \rightarrow \mathbb{R}$ are real-valued functions.

For $f, g \in H$, define $\langle f, g \rangle := \frac{1}{2\pi} \int_0^{2\pi} f(t) \overline{g(t)} dt$.

That is, if $f = f_1 + if_2$ and $g = g_1 + ig_2$ then $f(t) \overline{g(t)} = (f_1 g_1 + f_2 g_2) + i(-f_1 g_2 + f_2 g_1)$ and so

$$\frac{1}{2\pi} \int_0^{2\pi} f(t) \overline{g(t)} dt := \frac{1}{2\pi} \int_0^{2\pi} (f_1 g_1 + f_2 g_2) dt + \frac{i}{2\pi} \int_0^{2\pi} (-f_1 g_2 + f_2 g_1) dt.$$

Then $\langle \cdot, \cdot \rangle : H \times H \rightarrow \mathbb{C}$ is an inner product on H .

Example 9. : Let $S = \{ f_n(t) = e^{int} \mid n \in \mathbb{Z} \}$.

(1) If $n \neq m$, then $\langle f_n, f_m \rangle = \frac{1}{2\pi} \int_0^{2\pi} e^{i(n-m)t} dt = \left[\frac{1}{2\pi(m-n)} e^{i(m-n)t} \right]_0^{2\pi} = 0$.

(2) If $n = m$, then $\langle f_n, f_n \rangle = \frac{1}{2\pi} \int_0^{2\pi} 1 \cdot dt = \frac{2\pi}{2\pi} = 1$.

In consequence, $\langle f_n, f_m \rangle = \delta_{ij}$ and hence S is orthonormal.